When the Marriage Market Fails: How Changes in Relative Wages Shape Childbearing Decisions

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Abstract

Between 1980 and 2010 women's wages rose from under half to about two-thirds of men's, and a falling share of American women became mothers. Following Shenhav [2021], I measure each marriage market's female-to-male *potential* wage with a shift–share that interacts 1970 industry–occupation employment shares with national wage growth, and link it to Census and ACS data on women aged 22–44. A 10 percent increase in the relative potential wage lowers the probability that a woman has a child by 4.3 percentage points and delays her first birth by about a third of a year. Both effects hold in every specification I estimate, among women living in their state of birth, and at ages 40–44, where postponement cannot explain them; mothers do not have fewer children. The variation behind these estimates comes from men: where the relative potential wage rises, men's wages, employment and hours fall while women's do not, fewer women marry, and more mothers have never married. The evidence points to a failing marriage market rather than a higher price of women's time. Extending the data to 2024 shows the design's limits: after 2010 the shift–share no longer predicts actual relative wages, so the steep recent decline in motherhood lies outside what it can identify.

JEL codes: J12, J13, J16, J31. *Keywords:* fertility, childlessness, gender wage gap, marriage markets, shift–share.

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1 Introduction

Becker [1960, 1991] argued that marriage and families exist primarily to produce and raise children, with the gains from marriage coming from specialization and the exchange of household and market work. In the decades since, lifelong marriage combined with childrearing has stopped being the near-universal path to adulthood [Lundberg and Pollak, 2007]. Over the same decades women's earnings rose relative to men's. This paper asks whether the second change caused part of the first: how did the narrowing of the U.S. gender wage gap between 1980 and 2010 change whether, and when, women had children?

The two trends are large. Panel A of Figure 1 shows that the female-to-male wage ratio rose from below 50 percent in 1980 to about 65 percent by 2010, even after adjusting for education and occupation, while the total fertility rate stayed below replacement throughout. Panel B shows what lies underneath the fertility rate. Among women aged 22–44, the number of children per mother fell from about 2.6 in the 1960s to about 2.0 by 1990 and has barely moved since. After 1990, the decline in children per woman is a decline in the share of women who become mothers. Any account of recent American fertility has to explain why fewer women enter motherhood, and when those who do enter it.

Observed wages cannot answer the question, because fertility shapes women's labor supply and earnings as much as the reverse [Pollak, 2005, McNown and Rajbhandary, 2003]. I follow Shenhav [2021] and use a Bartik-style shift–share measure of the female-to-male *potential* wage in a marriage market, defined by state, education and race. The measure interacts each market's 1970 industry–occupation employment shares, separately for women and for men, with subsequent national wage growth in those cells computed outside the market's own state. It moves when national demand shifts toward the jobs that women in a market held in 1970, or away from the jobs the men held, and it does not respond to local fertility or marriage decisions. I link it to IPUMS microdata from the 1980, 1990 and 2000 Censuses and the 2010 American Community Survey and estimate reduced-form regressions with market, state-by-year, education-by-year, race-by-year and cohort-by-year fixed effects and race-by-state trends.

I study four outcomes that decompose fertility: whether a woman has at least one child, the number of children per mother, the number of children per woman (the product of the first two), and age at first birth. The main result is that relative potential wages act on entry and timing. A 10 percent increase lowers the probability that a woman has a child by 4.3 percentage points, against a mean of 65 percent, and raises the average age at first birth by 0.34 years. Children per mother does not fall. Children per woman falls by 0.07 in the baseline specification, as the decomposition implies, but that estimate is sensitive to the choice

of controls, so I treat it as supporting evidence. The entry and timing results are not: they keep their sign and significance when the market controls or trends are dropped, when the 2010 wave is excluded, when small cells are trimmed (which makes them larger), and when the sample is restricted to women living in the state where they were born. The entry effect is present at ages 40–44, where few first births remain, so it is not only postponement. The effects are not driven by pre-existing trends: future growth in the relative potential wage does not predict the change in motherhood or in children per woman between 1960 and 1970.

Who is responding, and why? The answer comes from asking what the instrument moves. A higher relative *potential* wage could reflect better prospects for women or worse prospects for men, and the two have different implications. If women's opportunities improve, the price of a mother's time rises and the classic substitution argument applies [Becker, 1960, Galor and Weil, 1993]. If men's prospects deteriorate, fewer men are attractive partners and fewer marriages form [Wilson, 1987, Autor et al., 2019]. In the data, the variation is overwhelmingly the second kind. Where the relative potential wage rises, men's observed wages fall sharply, their employment and hours fall, and women's wages, employment and hours do not change. In the same markets fewer women ever marry, married women are less likely to have children, never-married women's fertility is unchanged, and the share of mothers who have never married rises. This is the signature of a marriage market that is failing, the mechanism in the title, rather than of a rising price of women's time.

The last part of the paper asks whether the results extend to the present. Motherhood among women aged 22–44 fell much faster after 2010 than before it (Figure 1, Panel B), so it is natural to carry the design forward. I rebuild the shift–share from the raw 1970 Census and March CPS, extend it to 2024, and show that it stops working: its first stage on observed relative wages falls from an F-statistic of 29 over 1980–1990 to zero over 2010–2024. The national wage ratio has barely moved since 2010, and the measure's remaining movement no longer tracks actual wages. The post-2010 estimates therefore carry no causal content, and the decline in motherhood in that period lies outside what this design can explain. This mirrors what Shourian and Ghosh [2026] find for Shenhav's marriage results, and it is a practical caution for shift–share designs that rely on shares measured decades before the outcome.

The paper contributes to three literatures. First, it adds causal evidence on wages and fertility, where identification has been the central difficulty [Cai and Winters, 2024]. Previous work has used shocks to absolute male or female earnings [Kearney and Wilson, 2018, Autor et al., 2019, Amialchuk, 2013] or to observed relative incomes [Anelli et al., 2024]; I use relative potential wages and show that the response is on entry and timing rather than on family size. Second, it adds to work on the extensive margin of fertility and on childlessness [Baudin et al., 2015, Hazan and Zoabi, 2015, Doepke et al., 2023] by locating a cause of the fall in entry.

Third, it extends the marriage-market evidence of Shenhav [2021] and Autor et al. [2019] from marriage to births and shows that the same variation that lowers marriage lowers motherhood and shifts it outside marriage. The fertility response is not only a mechanical consequence of fewer marriages, however, since married women in the affected markets also have fewer children.

The paper proceeds as follows. Section 2 reviews related work. Section 3 sets out the two channels and what distinguishes them in the data. Section 4 describes the data and the empirical strategy. Section 5 presents the main results, their robustness and their heterogeneity. Section 6 addresses measurement, pre-trends and migration. Section 7 presents the mechanism evidence. Section 8 carries the design to 2024. Section 9 concludes.

2 Related Literature

Wages and fertility. The economic analysis of fertility starts from Becker [1960] and Willis [1973], in which children are time-intensive and their price is the value of the time of whoever cares for them. Because that person has mostly been the mother, a rise in women's wages raises household income and the price of a child at the same time. Galor and Weil [1993] sharpen this into a prediction about relative wages: when women's wages rise relative to men's, the price of children rises faster than household resources and fertility falls. Siegel [2017] argues that the narrowing gender gap, rather than the level of women's wages, drives the long-run U.S. fertility trends, and Day [2018] show that growth favoring women can lower fertility initially and raise it later once family policy adapts.

Empirically, observed wages are endogenous to fertility [Millimet, 2000, McNown and Rajbhandary, 2003, Pollak, 2005], and recent work uses shocks to gender-specific labor demand. Kearney and Wilson [2018] find that fracking booms, which raised men's earnings, increased both marital and nonmarital births without raising marriage. Autor et al. [2019] show that trade shocks to men's relative earnings reduced marriage and fertility and raised the share of children born to unmarried mothers, while shocks concentrated in female-intensive industries had smaller effects of the opposite sign. Anelli et al. [2024] find that automation, which lowered men's relative standing, shifted births from inside to outside marriage. Cai and Winters [2024] use industry- and major-specific earnings as imperfect instruments and find modest negative effects of women's income on fertility among college graduates. Amialchuk [2013] find that higher male earnings raise fertility among young married women. A theme across these papers is that men's and women's earnings work through different channels, and that men's earnings work partly through marriage.

Relative potential wages and marriage. Shenhav [2021] builds a shift–share measure of the female-to-male potential wage, following the logic of Bartik [1991] as used by Aizer [2010] and Bertrand et al. [2015], and shows that a higher relative wage lowers marriage and raises the never-married share. I use her measure and her identifying variation and ask about births rather than marriage. The modern shift–share literature emphasizes that identification comes either from the shares or from the shocks [Goldsmith-Pinkham et al., 2020, Borusyak et al., 2022] and that inference needs care [Adão et al., 2019]; I return to these points in Section 4.

Margins of fertility. The demographic record increasingly points to entry into motherhood rather than family size as the margin that moves. Baudin et al. [2015] document that childlessness in the United States is large and U-shaped in education, with different causes at the top and the bottom. Hazan and Zoabi [2015] show that the education gradient in family size flattened once educated women could buy substitutes for their own time. Caucutt et al. [2002] model postponement and completed fertility as distinct decisions, and Kearney et al. [2022] document that the post-2007 decline in U.S. births is broad and not explained by economic conditions or policy. I add causal evidence that relative wages moved entry and timing, and evidence on why.

The household. Children's time cost has stayed with mothers [Aguiar and Hurst, 2007, Guryan et al., 2008, Ramey and Ramey, 2010, Kleven et al., 2019, 2021, Cortés and Pan, 2023], and the decline of household specialization [Lundberg and Pollak, 2007, Lundberg et al., 2016] and the structure of work [Goldin, 2006, 2014] shape what mothers can do. McDonald [2000] argues that fertility stays low where equity in the labor market outruns equity at home. These arguments concern the price of a mother's time; Section 7 shows that the variation used here does not raise that price, which is why the paper's mechanism runs through the marriage market instead.

3 Two Ways a Relative Wage Can Move Fertility

A higher female-to-male potential wage can come about in two ways, and each has its own route to fertility.

The price of a mother's time. If women's own prospects improve, the time a mother spends with a child costs more in forgone earnings. In the New Home Economics the substitution effect of a higher female wage dominates its income effect when the female wage rises relative to the male wage [Becker, 1960, Willis, 1973, Galor and Weil, 1993]. Higher-wage women then have fewer children or none [Baudin et al., 2015], and they postpone the

first birth to protect early-career earnings [Miller, 2011]. This channel predicts that a higher relative potential wage raises women's observed wages, employment and hours.

The marriage market. If men's prospects deteriorate, the gains from marriage fall. A woman's threshold for an acceptable partner rises relative to what the men around her can offer, and fewer marriages form [Shenhav, 2021]. The argument goes back to Wilson [1987]'s "marriageable men" and has been documented for trade shocks [Autor et al., 2019]. Because most births in this period took place within marriage, fewer and later marriages mean fewer and later first births. Some births move outside marriage, which raises the share of mothers who have never married [Kearney and Wilson, 2018, Autor et al., 2019]. Within marriage, lower husbands' earnings lower fertility through an income effect. This channel predicts that a higher relative potential wage lowers men's observed wages and employment, lowers marriage, and raises the never-married share of mothers.

Both channels predict less and later childbearing, so the fertility estimates alone cannot separate them. Their predictions for the labor market differ, though, and Section 7 tests them directly. Both can also operate at once, and the margins are not separately identified as *channels*: marriage, labor supply and fertility are all outcomes of the same shock, so conditioning on one to isolate another conditions on a variable the treatment moves. I therefore read the mechanism evidence through what the instrument moves, rather than through a formal decomposition of the fertility effect.

4 Data and Empirical Strategy

4.1 Matching markets and the relative potential wage

A matching market μ groups individuals who live in the same state s and share an education level e (high school or less; some college or more) and a race and ethnicity r (non-Hispanic White, non-Hispanic Black, Hispanic). This follows Shenhav [2021] and Bertrand et al. [2015] and is meant to capture the boundaries within which most people meet partners and form families. There are 306 markets.

The potential wage of sex g in market μ and year t follows Shenhav [2021]:

$$\widehat{w}_{\mu gt} = \sum_{j} \frac{E_{j\mu g,1970}}{E_{\mu g,1970}} \sum_{o} \left(\frac{E_{oj\mu g,1970}}{E_{j\mu g,1970}} \pi^{W*}_{ojt,-s} \right) w_{ojt,-s},$$

where $E_{j\mu g,1970}/E_{\mu g,1970}$ is the 1970 share of group (μ, g) employed in industry j,

$E_{oj\mu g,1970}/E_{j\mu g,1970}$ the 1970 share in occupation o within that industry, $\pi^{W*}_{ojt,-s}$ updates the within-industry occupation mix with national changes, and $w_{ojt,-s}$ is the national mean log wage in the occupation–industry cell, computed excluding state s. The industry and occupation groupings are listed in Appendix Table A3. The regressor is the log ratio $\ln(\widehat{w}_{\mu,\text{female},t}/\widehat{w}_{\mu,\text{male},t})$. The shares are fixed in 1970, a decade before the first outcome year, and the wage growth that moves them is national and excludes the market's own state.

I use the measure exactly as distributed by Shenhav [2021], built from the 1970 Census (1 percent) and the 1980–2011 March CPS, for 1980, 1990, 2000 and 2010. Section 8 uses a version that I rebuild from the raw data and extend to 2024. Appendix Table A2 compares the two on the years they share.

4.2 Fertility data and outcomes

Fertility outcomes come from the 1980, 1990 and 2000 Census 5 percent samples and the 2010 ACS, accessed through IPUMS USA [Ruggles et al., 2025]. I keep women aged 22–44, old enough that most have finished school and young enough to be at risk of a first birth, and assign them to markets by state of residence, education and race. I collapse the data to cells defined by market, birth cohort, age band and year; the analysis sample contains 27,228 cells and about 6.6 million women (Appendix Table A1).

The four outcomes are: (i) the share of women with at least one own child in the household; (ii) the number of children per mother, the cell's total children divided by its number of mothers; (iii) the number of children per woman, which equals (i) times (ii); and (iv) age at first birth, the mother's age minus the age of her eldest co-resident child, defined for mothers only. Outcomes (i)–(iii) decompose completed fertility into entry and family size. Because the Census and ACS count only children living with their mother, all four are co-resident measures; Section 6 compares them with children ever born, which the 1980 and 1990 Censuses record.

4.3 Estimating equation

Following Shenhav [2021], who builds on Aizer [2010] and Bertrand et al. [2015], I estimate reduced-form regressions of fertility on the relative potential wage:

$$Y_{\mu ct} = \beta \ln\left(\frac{\widehat{w}_{\mu,\text{female},t}}{\widehat{w}_{\mu,\text{male},t}}\right) + \alpha_\mu + \delta_{rt} + \chi_{et} + \gamma_{st} + \xi_{ct} + \rho_{rs}\cdot t + X_{\mu t}\Phi + \nu_{\mu ct}, \tag{1}$$

where $Y_{\mu ct}$ is an outcome for women of cohort c in market μ in year t. Market fixed effects α_μ absorb permanent differences across markets. Race-by-year, education-by-year and state-by-year effects absorb national trends by group and every state-level shock, including state policy and state business cycles. Cohort-by-year effects absorb differences in age composition, and race-by-state linear trends allow smooth differential trends. $X_{\mu t}$ contains the average years of schooling of women and of men and the sex ratio in the market, as in Shenhav [2021]. Identification therefore comes from differences in the growth of the relative potential wage across education and race groups within the same state and year. Regressions are weighted by the female population of the cell (by the number of mothers for children per mother), and standard errors are clustered by state. Outcomes are divided by ten, so β is the effect of a 10 percent increase in the relative potential wage.

What the variation is. Equation (1) is a reduced form, so its coefficients do not need a strong first stage to be estimated consistently. They do need one to be read as effects of relative *wages*. Over 1980–2010 the measure predicts observed relative wages: in the March CPS, using Shenhav's first-stage specification, a one-log-point increase raises the observed female-to-male log wage by 0.59 ($F = 9.6$; Table 8, Panel A), and in hourly wages from the 1980–2000 Censuses by 0.53 ($F = 14.7$; Table 6). Section 7 shows which wage does the moving, and Section 8 shows that the first stage disappears after 2010, which is why the main analysis stops there.

5 Results

5.1 Main estimates

Table 1 reports the main estimates. In Panel A, a 10 percent increase in the female-to-male potential wage lowers the probability that a woman has at least one child by 4.3 percentage points (column 1), against a mean of 64.9 percent. It raises the average age at first birth by 0.34 years, about four months (column 4), against a mean of 23.7 years. Children per mother does not fall: the estimate is +0.027 and insignificant (column 2). Children per woman falls by 0.071 (column 3), which is what the decomposition implies: $-0.0432 \times 2.072 + 0.0273 \times 0.649 = -0.072$. The fall in completed fertility is therefore entirely a fall in entry. Women in markets where the relative wage rose became mothers less often and later, and those who became mothers had as many children as before.

Column 2 should be read with care. It conditions on motherhood, which the treatment moves. If the women deterred from motherhood are those who would have had small families,

the average among the remaining mothers rises even if no woman changes her plans. Columns 1 and 3 condition on nothing and do not have this problem. Age at first birth is defined only for mothers and shares it in principle; the age profile below shows that the delay is largest among the oldest women, whose fertility is nearly complete.

Panel B adds the average (sex-combined) potential wage, so the relative wage is estimated holding the level of potential earnings in the market constant. The relative-wage coefficients barely change: -0.042 on entry and $+0.34$ years on age at first birth. The average wage enters with the opposite sign on entry ($+0.050$): when both men's and women's prospects improve, more women become mothers, consistent with an income effect [Becker, 1960, Schaller, 2016, Kearney and Wilson, 2018]. It also raises age at first birth and lowers children per mother. What lowers fertility is a worsening of men's prospects *relative to* women's, not a change in the level of earnings.

5.2 Robustness

Table 2 varies one element of the specification at a time. Two things stand out.

First, the market controls are themselves outcomes. Regressed on the relative potential wage with the full structure of equation (1), a 10 percent increase lowers average years of schooling by 0.35 years for women and 0.38 years for men, while the sex ratio does not respond (table notes). Dropping the schooling controls (rows 2 and 3) leaves the entry effect unchanged at -0.043 and the delay at $+0.22$ years. It moves children per mother up to $+0.07$ and children per woman to -0.035, which is no longer significant. Children per woman is therefore the one outcome whose estimate depends on a debatable control, and I do not rely on it.

Second, the entry and timing results survive every variation. Without the race-by-state trends (row 4), without any controls or trends (row 5), and without cohort-by-year effects (row 6), entry falls by between 2.9 and 5.0 percentage points and age at first birth rises by between 0.28 and 0.35 years, all significant at 1 percent. Dropping the 2010 wave (row 7) gives -0.027 and $+0.23$. Dropping cells with fewer than 50 sampled women (row 8) *raises* both, to -0.059 and $+0.51$, which is what classical noise in small cells produces and the opposite of what a result manufactured by thin cells would do.

5.3 Heterogeneity

Table 3 reports heterogeneity by age, education and race. For education and race I interact the relative wage with group indicators in a single regression, so that the fixed effects of equation (1) are estimated on all cells. Separate regressions by group are not informative here, because within a single education or race group several fixed effects absorb the variation that identifies the pooled estimate.

Age. The entry effect is negative at every age (Panel A). It is smallest among women aged 22–29 (-1.8 points) and largest at 30–39 (-5.6 and -6.2 points), and it remains at -3.9 points among women aged 40–44, for whom few first births remain. Children per woman falls most at 36–44 (-0.087 and -0.070). If higher relative wages only postponed motherhood, the entry effect would vanish at the oldest ages. It does not, so part of the response is forgone motherhood, not just delayed motherhood. The delay in first births grows with age, from $+0.11$ years at 22–29 to $+0.65$ years at 40–44. Among younger women a later first birth may not yet have happened, so the oldest group is the cleanest measure of the delay.

Education. The entry effect is larger for women with some college (-6.5 points) than for women with a high school education or less (-3.5 points; $p = 0.08$ for equality). The delay is concentrated among less-educated women ($+0.53$ years, against an insignificant -0.19 for the more educated; $p < 0.001$). Children per woman falls significantly only for the less educated (-0.083). Among women with some college, children per mother rises ($+0.096$), which is the composition effect described above.

Race and ethnicity. The entry effect is largest for White women (-6.3 points) and smaller for Black women (-1.8 points; $p < 0.001$ for equality), and it is absent for Hispanic women. Black women respond more on timing, with first births delayed by 0.62 years against 0.22 for White and 0.24 for Hispanic women, and children per woman falls for both Black and White women (-0.089 and -0.072).

6 Measurement, Pre-Trends, and Migration

6.1 Co-resident children and children ever born

The Census and ACS count only children living with their mother. A woman whose children have left home looks childless, and her eldest co-resident child may not be her first. The 1980 and 1990 Censuses also record the number of children ever born, which allows a direct check. Table 4 estimates equation (1) on the 1980 and 1990 waves for both measures, age band

by age band. With only two waves the design has little power for the pooled effect, which is insignificant on both measures. The question here is whether the two measures move together. They do: at every age the two estimates have the same sign and similar size. At 30–35, where most women have had their first child and few children have left home, both measures show a significant decline in motherhood (-3.5 and -5.0 points). At 36–39, the decline in children per woman is larger on the direct measure (-0.21) than on the co-resident one (-0.05), as expected when older children have begun to leave home. The co-resident measures therefore understate rather than create the fertility response.

6.2 Pre-trends

If markets with larger future gains in the relative potential wage were already on different fertility paths, the estimates would pick up those paths. Panel A of Table 5 regresses the change in fertility between 1960 and 1970, a decade before the sample begins, on the change in the relative potential wage over 1980–2010. Future relative-wage growth does not predict past changes in motherhood or in children per woman, either with or without state fixed effects. The coefficient on the change in age at first birth is negative and significant at 10 percent only with state fixed effects. That is the opposite sign to the main result, so any pre-trend would work against the delay.

6.3 Migration

Women may move toward markets with better prospects, and if movers differ in fertility, the composition of a market could respond to the relative wage. Panel B of Table 5 splits each cell by whether the woman lives in her state of birth and interacts the relative wage with an indicator for movers. Among women living in their state of birth, who have not chosen their market by moving, the effects are as large as in the full sample: -4.5 percentage points on entry and $+0.38$ years on age at first birth. The additional effect for movers is small and insignificant for both outcomes. Selective migration does not drive the results.

7 Why: The Marriage Market

7.1 What a rise in the relative potential wage looks like in the labor market

Section 3 showed that the two channels make opposite predictions about which wages and whose work respond. Table 6 tests them. Panel A regresses observed log hourly wages in each market and year on the relative potential wage, using the 1980–2000 Censuses, which report the weeks worked needed to compute an hourly wage. A one-log-point increase raises the observed female-to-male wage ratio by 0.53 (the first stage), but it does so almost entirely by lowering men's wages: the male log wage falls by 0.64, while the female log wage does not change significantly (-0.11). Panel B shows the same split in work. A 10 percent increase in the relative potential wage lowers men's employment by 4.6 percentage points and their usual weekly hours by 1.7. Women's employment and hours do not change.

This is the key fact for interpretation. Over 1980–2010, the markets in which the female-to-male potential wage rose were markets in which the men's jobs lost ground. This is consistent with the decline in demand for the jobs in which many men worked in 1970 [Autor et al., 2019]; women's prospects did not improve differentially in the same markets. The variation does not raise the price of women's time, and the substitution channel cannot be what drives the fertility response.

7.2 Marriage and births outside marriage

Table 7 follows the marriage-market channel through to family formation. A 10 percent increase in the relative potential wage lowers the share of women who have ever married by 5.8 percentage points and the share currently married with a spouse present by 6.0 points (columns 1 and 2), confirming Shenhav [2021] in the Census and ACS cells used here. Column 5 shows that the share of mothers who have never married rises by 2.8 percentage points, against a mean of 10 percent. Births move outside marriage, as in Autor et al. [2019] and Anelli et al. [2024].

Columns 3 and 4 show where the decline in motherhood comes from. Among ever-married women, motherhood falls by 3.7 percentage points. Among never-married women it does not change (-0.1 points). Both columns condition on marital status, which the treatment moves, so they are descriptive. They still rule out one reading: the fall in motherhood is not only an arithmetic consequence of fewer marriages, since married women in the affected markets

also have fewer children. That is what the marriage-market channel predicts when husbands' earnings fall: Kearney and Wilson [2018] and Amialchuk [2013] find that men's earnings raise births within marriage, so lower men's earnings should lower them. Never-married women's fertility, by contrast, is unaffected, so as marriages decline, a larger share of births occurs outside marriage.

Panel B adds the average potential wage. The relative-wage effects are unchanged, and the average wage *raises* marriage (+9.4 points) and motherhood within marriage. Marriage and births respond to men's prospects relative to women's, not to the general level of earnings.

7.3 Interpretation

Taken together, these results describe a marriage market that fails rather than a mother's time that becomes too expensive. When men's labor market prospects deteriorate relative to women's, fewer women marry, married women have fewer children, first births come later, and a larger share of births occur outside marriage. These are the outcomes in Tables 1 and 7. Women's own wages and labor supply do not move, so the substitution channel of Section 3 plays little role in this variation. This does not mean women's wages are irrelevant to fertility. It means that between 1980 and 2010 the gender wage gap narrowed in large part because men's wages fell [Autor et al., 2019], and it was this part of the convergence that the shift–share isolates and that reduced childbearing.

8 Carrying the Design to 2024

Motherhood fell much faster after 2010 than before. Among women aged 22–44 in my cells, the share with a co-resident child fell from 67.6 percent in 1980 to 62.0 percent in 2010, a decline of about 6 points in three decades, and then to 50.6 percent in 2024, a further 11 points in fourteen years. Children per mother was flat at about 2.07 from 2000 on (Figure 1, Panel B). Since the post-2010 decline is again a decline in entry, it is natural to ask whether relative wages account for it.

To answer, I rebuild the relative potential wage from the raw 1970 Census and March CPS, following Shenhav [2021]'s construction, and extend it to the 2015, 2017, 2019, 2022 and 2024 ACS waves. Over 1980–2010 the rebuilt measure correlates 0.99 with hers in levels, but after the fixed effects of equation (1) the correlation falls to 0.78. As classical measurement error predicts, the rebuilt measure gives smaller coefficients on the same cells: −0.030 on entry and +0.11 years on age at first birth, against −0.043 and +0.34 (Appendix Table A2).

Table 8 and Figure 4 show what happens as the window is extended. Panel A reports the first stage: the observed female-to-male log wage gap in the March CPS regressed on the relative potential wage, with Shenhav's specification. The relationship is strong over 1980–1990 (coefficient 1.28, $F = 29$), weakens as later decades are added (0.87 over 1980–2000, 0.59 over 1980–2010), and is zero over 2010–2024 (0.00, $F = 0.0$). The measure has not stopped moving: its variation within markets after the fixed effects is no smaller after 2010 than before. What has changed is that this variation no longer tracks actual wages. The shares are now fifty years old, the national wage ratio has barely moved since 2010, and the occupational structure the measure was built on has changed.

Panel B reports the fertility estimates on the rebuilt measure. The entry effect is negative and significant at 10 percent or better over 1980–2000, 1980–2010 and 1980–2024, and zero over 2000–2024 and 2010–2024; over 1980–1990 alone, with only two waves, it is imprecise. The estimate for age at first birth remains positive after 2010. Without a first stage, however, neither the null nor the positive estimate can be read as an effect of relative wages: the regressor no longer measures the thing it is meant to measure. Pooling 1980–2024 averages a period with a first stage and a period without one and has no clean interpretation either.

The conclusion for this paper is that the causal evidence covers 1980–2010, and that the steep post-2010 decline in motherhood lies outside what this design can explain. The national female-to-male wage ratio was close to flat after 2010 (Figure 1, Panel A), so relative wages are in any case unlikely to account for much of that decline. Other explanations are more plausible, including the Great Recession and housing costs, the fall in teen and unintended births, and changing priorities across cohorts [Kearney et al., 2022]. The broader lesson concerns shift–share designs. Shares measured in 1970 predicted relative wages well for two decades and not at all by the 2010s. A design validated on one period cannot be assumed to work on another without re-establishing its first stage, a point that Shourian and Ghosh [2026] also make for Shenhav's marriage results.

9 Conclusion

I study how changes in the female-to-male potential wage shaped childbearing in the United States between 1980 and 2010, using the shift–share measure of Shenhav [2021] and Census and ACS microdata for women aged 22–44.

Three results emerge. First, relative potential wages act on entry into motherhood and on its timing, not on family size. A 10 percent increase lowers the probability that a woman has a child by 4.3 percentage points and delays the first birth by about a third of a year, while

mothers do not have fewer children. Both effects survive every specification I estimate, are not explained by pre-existing trends or by migration, and the entry effect persists at ages 40–44, so it reflects forgone as well as postponed motherhood. Second, the variation that produces these effects is a decline in men's prospects relative to women's: men's wages, employment and hours fall, women's do not, marriage declines, married women have fewer children, and births move outside marriage. The evidence points to a failing marriage market rather than a higher price of women's time. Third, the design has a limited shelf life. After 2010 the shift–share no longer predicts actual relative wages, so it cannot speak to the sharp decline in motherhood since then.

The study has limits. The outcomes are based on co-resident children. The comparison with children ever born suggests this understates the response rather than creating it, but direct birth records would be better. The estimates for children per mother and children per woman depend on controls that themselves respond to the treatment, so I treat them as supporting evidence. And the mechanism evidence shows what the instrument moves, not a formal decomposition of the fertility effect into channels.

For policy, the results suggest that the link between gender wage convergence and low fertility in this period ran less through women's careers than through men's. When men's labor market position deteriorates, fewer couples form, and those that form have children later and less often. Policies that address the earnings prospects of less-educated men and the economic foundations of partnership may matter for fertility as much as policies that lower the cost of a mother's time.

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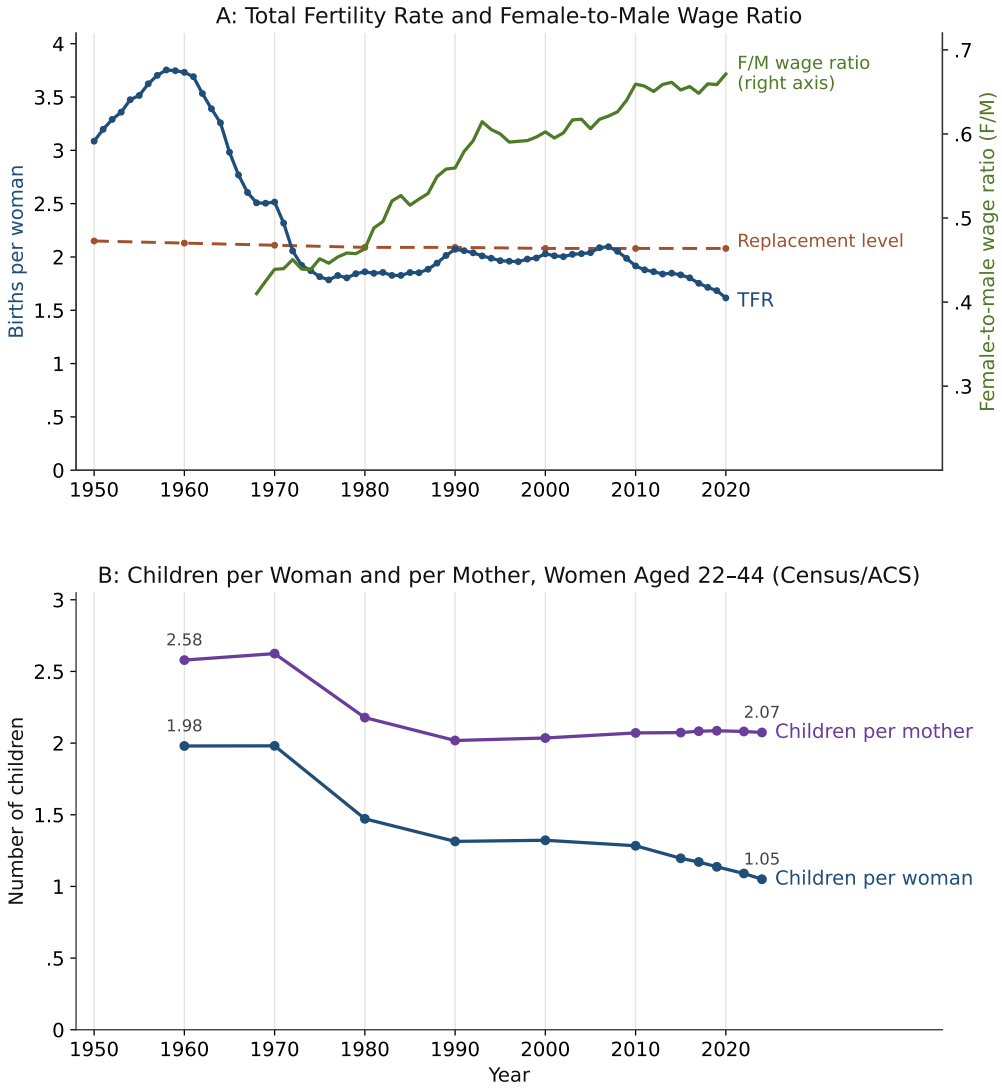
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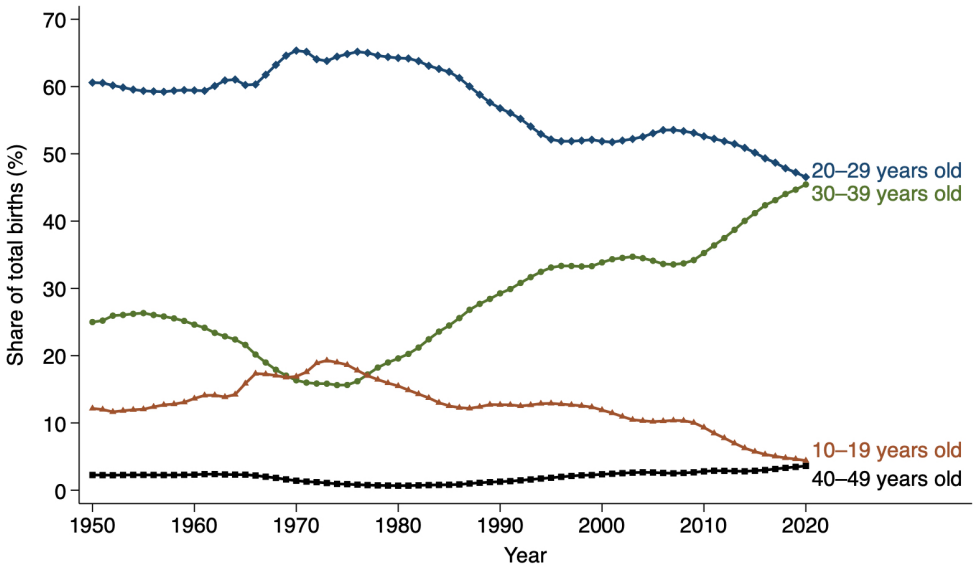
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Figure 1: Fertility and the Female-to-Male Wage Ratio, United States



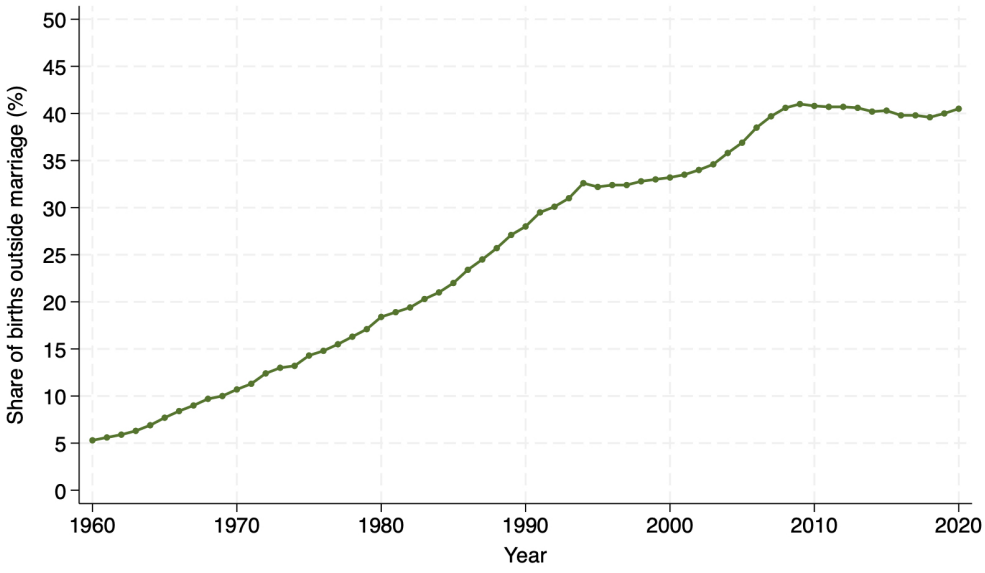
Note: Panel A plots the total fertility rate (TFR, left axis) against the replacement level, 1950–2020, and the female-to-male wage ratio (right axis), 1968–2020. The wage ratio is the exponentiated coefficient on a female indicator from year-by-year regressions of log wages on the indicator with education and occupation fixed effects, for workers aged 18–64 with positive hours and earnings (March CPS). TFR sources: Human Fertility Database [2024], Organisation for Economic Co-operation and Development (OECD) [2024], Gietel-Basten and Scherbov [2020], processed by Dattani et al. [2025]. Panel B plots the average number of own children in the household among all women aged 22–44 (children per woman) and among those with at least one child (children per mother); children per woman equals the share of women who are mothers times children per mother. IPUMS 1960–2000 Censuses and 2010–2024 ACS.

Figure 2: Share of Births by Age of Mother, United States



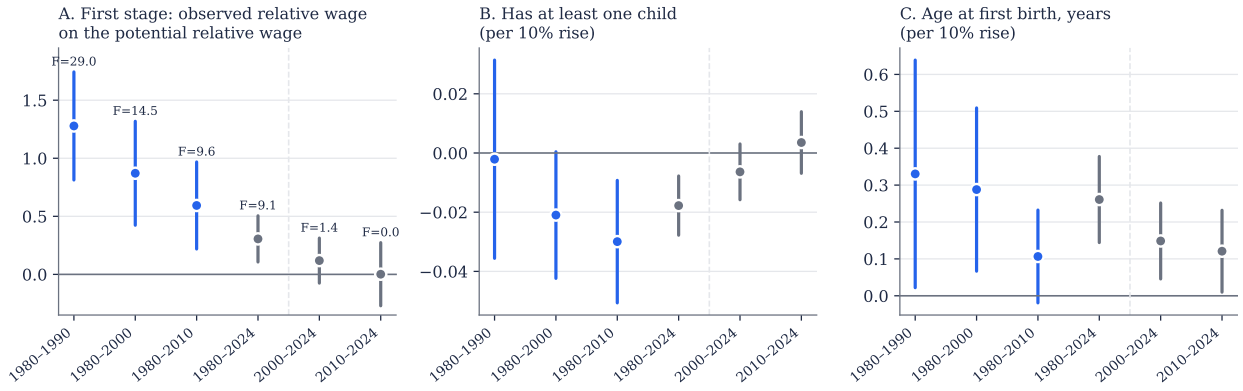
Note: Estimated share of all births by age of the mother, 1950–2020. Source: United Nations, Department of Economic and Social Affairs, Population Division [2024], processed by Dattani et al. [2025].

Figure 3: Share of Children Born Outside Marriage, United States



Note: Share of all births to mothers who were not married at the time of birth, 1960–2020. Source: Organisation for Economic Co-operation and Development (OECD) [2024], processed by Dattani et al. [2025].

Figure 4: The First Stage and the Fertility Estimates, by Window



Note: Point estimates and 95 percent confidence intervals from Table 8. Panel A: coefficient from regressing the observed female-to-male log wage gap (March CPS) on the relative potential wage, with F-statistics. Panels B and C: equation (1) on the rebuilt and extended relative potential wage, per 10 percent increase. Blue markers are windows that end by 2010; gray markers include post-2010 waves. Standard errors clustered by state.

Table 1: The Effect of Relative Potential Wages on Fertility, 1980–2010

	(1) Has at least one child	(2) Children per mother	(3) Children per woman	(4) Age at first birth
Panel A. Relative potential wage only				
Relative potential wage	-0.043***	0.027	-0.071**	0.337***
	(0.012)	(0.022)	(0.030)	(0.088)
Mean of outcome	0.649	2.072	1.345	23.651
Cells	24,696	24,307	24,696	24,284
Panel B. Adding the average (sex-combined) potential wage				
Relative potential wage	-0.042***	0.026	-0.070**	0.343***
	(0.008)	(0.020)	(0.028)	(0.097)
Average potential wage	0.050***	-0.069***	0.032	0.193*
	(0.011)	(0.023)	(0.040)	(0.103)

Notes: Cells are education × race × state × birth cohort × year for women aged 22–44 in the 1980, 1990 and 2000 Censuses and the 2010 ACS. The relative wage is the log female-to-male potential wage of Shenhav [2021]. All regressions include matching-market, education-by-year, race-by-year, state-by-year and birth-cohort-by-year fixed effects, race-by-state linear trends, and the average years of schooling of women and of men and the sex ratio in the market, unless stated otherwise. Outcomes are divided by ten, so each coefficient is the effect of a 10 percent increase in the relative potential wage. Regressions are weighted by the female population of the cell (by the number of mothers in column 2). Standard errors clustered by state in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column 3 equals column 1 times column 2 in levels; the implied effect (-0.0718) matches the direct estimate (-0.0713). Column 2 conditions on motherhood, which the relative wage itself moves, and is descriptive.

Table 2: Robustness of the Main Estimates, 1980–2010

	(1) Has at least one child	(2) Children per mother	(3) Children per woman	(4) Age at first birth
(1) Baseline, equation (1)	-0.043***	0.027	-0.071**	0.337***
	(0.012)	(0.022)	(0.030)	(0.088)
(2) Sex ratio only (no schooling controls)	-0.043***	0.072***	-0.035	0.219***
	(0.011)	(0.024)	(0.027)	(0.079)
(3) No market controls	-0.043***	0.073***	-0.035	0.219***
	(0.011)	(0.026)	(0.028)	(0.080)
(4) No race-by-state trends	-0.029***	0.018	-0.048*	0.349***
	(0.010)	(0.022)	(0.029)	(0.092)
(5) No controls and no trends	-0.029***	0.075**	-0.007	0.278***
	(0.010)	(0.031)	(0.031)	(0.085)
(6) No cohort-by-year effects	-0.050***	0.019	-0.083*	0.287***
	(0.017)	(0.023)	(0.045)	(0.075)
(7) 1980–2000 only	-0.027**	0.024	-0.027	0.229***
	(0.011)	(0.023)	(0.036)	(0.079)
(8) Cells with at least 50 women	-0.059***	0.031	-0.100**	0.512***
	(0.020)	(0.024)	(0.048)	(0.092)

Notes: Specification, weights and clustering as in Table 1: equation (1) on 1980–2010 cells with Shenhav's relative potential wage; coefficients are per 10 percent increase; standard errors clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Each row changes one element of the baseline specification. Rows (2), (3) and (5) answer the concern that the schooling controls are themselves outcomes: regressed on the relative potential wage with the full structure of equation (1), a 10 percent increase lowers the average years of schooling of women by 0.35 (s.e. 0.09) and of men by 0.38 (s.e. 0.11), while the sex ratio does not respond (+0.001, s.e. 0.035). Row (7) drops the 2010 ACS wave; row (8) drops cells with fewer than 50 sampled women.

Table 3: Heterogeneity by Age, Education, and Race, 1980–2010

	(1) Has at least one child	(2) Children per mother	(3) Children per woman	(4) Age at first birth
Panel A. By age (separate regressions)				
Ages 22–29	-0.018*	0.067**	0.010	0.106
	(0.011)	(0.031)	(0.027)	(0.089)
Ages 30–35	-0.056***	0.085***	-0.048	0.202*
	(0.017)	(0.032)	(0.040)	(0.117)
Ages 36–39	-0.062***	0.056	-0.087**	0.555***
	(0.016)	(0.034)	(0.043)	(0.169)
Ages 40–44	-0.039**	-0.008	-0.070*	0.648***
	(0.016)	(0.040)	(0.042)	(0.194)
Panel B. By education (interactions, common fixed effects)				
High school or less	-0.035***	0.003	-0.082***	0.528***
	(0.011)	(0.024)	(0.032)	(0.100)
Some college or more	-0.065***	0.096***	-0.041	-0.185
	(0.020)	(0.032)	(0.047)	(0.150)
p-value, equal effects	0.083	0.014	0.387	0.000
Panel C. By race and ethnicity (interactions, common fixed effects)				
Non-Hispanic White	-0.063***	0.067***	-0.072**	0.217**
	(0.010)	(0.020)	(0.034)	(0.098)
Non-Hispanic Black	-0.018**	-0.034	-0.089***	0.622***
	(0.009)	(0.026)	(0.029)	(0.101)
Hispanic	0.003	-0.019	-0.014	0.235*
	(0.012)	(0.034)	(0.043)	(0.123)
p-value, White = Black	0.000	0.000	0.562	0.000

Notes: Specification, weights and clustering as in Table 1: equation (1) on 1980–2010 cells with Shenhav's relative potential wage; coefficients are per 10 percent increase; standard errors clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel A estimates equation (1) separately for each age band. Panels B and C interact the relative potential wage with group indicators in a single regression on all cells, so the fixed effects are estimated on the full sample; separate regressions by education or race are not used because, within one education or race group, several of the fixed effects in equation (1) absorb the variation that identifies the pooled estimate.

Table 4: Co-Resident Children versus Children Ever Born, 1980 and 1990

	Has at least one child		Children per woman	
	Co-resident	Ever born	Co-resident	Ever born
	(1)	(2)	(3)	(4)
Ages 22–29	0.025	0.016	0.120**	0.097**
	(0.023)	(0.018)	(0.047)	(0.045)
Ages 30–35	-0.035**	-0.050***	-0.027	-0.067
	(0.016)	(0.017)	(0.055)	(0.052)
Ages 36–39	-0.024	-0.034*	-0.048	-0.206**
	(0.016)	(0.019)	(0.082)	(0.092)
Ages 40–44	0.004	0.004	-0.041	-0.090
	(0.024)	(0.018)	(0.097)	(0.113)
All ages 22–44	0.004	-0.007	0.032	-0.065
	(0.014)	(0.011)	(0.047)	(0.052)

Notes: Columns 1 and 3 use the co-resident measures of the main analysis; columns 2 and 4 use the number of children ever born (IPUMS CHBORN), which counts children who have left home and is available only in the 1980 and 1990 Censuses. Same specification as equation (1) on the 1980 and 1990 waves only. With two waves the design has little power for the pooled effect; the table is a test of the *measure*, asking whether the two counts move together with the relative wage, not a second estimate of the main effect. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Pre-Trends and Migration

	(1) Has at least one child	(2) Children per mother	(3) Children per woman	(4) Age at first birth
Panel A. Placebo: 1960–1970 change in fertility on the 1980–2010 change in the relative wage				
Without state fixed effects	-0.015	–	0.006	-0.180
	(0.013)		(0.101)	(0.154)
With state fixed effects	-0.001	–	0.075	-0.320*
	(0.022)		(0.084)	(0.184)
Panel B. Women living in their state of birth versus movers				
Relative wage (non-movers)	-0.045***	0.017	-0.086**	0.384***
	(0.013)	(0.022)	(0.036)	(0.106)
× born in another state or abroad	0.006	0.035*	0.046*	0.030
	(0.006)	(0.020)	(0.027)	(0.058)

Notes: Panel A regresses the 1960–1970 change in each outcome (1960 and 1970 Censuses, women aged 22–44, cells by market and age band) on the change in the relative potential wage over 1980–2010, with education-by-race-by-age-band fixed effects, without and with state fixed effects; the main specification's state-by-year effects correspond to the second row. Children per mother is not computed for the placebo. Panel B splits each cell by whether the woman lives in her state of birth and interacts the relative wage with an indicator for movers (born in another state or abroad), with the full specification of equation (1) plus mover-by-year effects. Standard errors clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: What a Rise in the Relative Potential Wage Looks Like in the Labor Market

	(1) Log relative wage	(2) Log female wage	(3) Log male wage
Panel A. Observed hourly wages, 1980–2000 (coefficient per log point of the instrument)			
Relative potential wage	0.529***	-0.107	-0.636***
	(0.138)	(0.177)	(0.207)
Cells	771	771	771

Panel B. Employment and hours, ages 22–44, 1980–2010 (per 10% rise)	Employed	Usual weekly hours
Men	-0.046***	-1.675***
	(0.013)	(0.625)
Mean	0.827	39.25
Women	-0.005	-0.001
	(0.014)	(0.510)
Mean	0.683	28.48

Notes: Panel A regresses the observed mean log hourly wage of women, of men, and their difference in each matching market and year (1980, 1990 and 2000 Censuses; workers aged 22–44 with positive wage income who worked at least 27 weeks and 20 hours a week; markets with at least 25 workers of each sex; the 2010 ACS reports weeks worked only in intervals and is excluded) on the relative potential wage, with matching-market, education-by-year, race-by-year and state-by-year fixed effects, race-by-state trends and the market controls; coefficients are per log point, so column 1 is the first stage. Panel B uses cells of men and women aged 22–44 by birth cohort and the full specification of equation (1); coefficients are per 10 percent increase. Weighted by population; standard errors clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: The Marriage Market, 1980–2010

	(1)	(2)	(3)	(4)	(5)
	Ever	Married,	Has at least one child		Never-married
	married	spouse present	Ever married	Never married	share of mothers
Panel A. Relative potential wage only					
Relative potential wage	-0.058***	-0.060***	-0.037***	-0.001	0.028***
	(0.016)	(0.017)	(0.011)	(0.028)	(0.009)
Mean of outcome	0.608	0.593	0.793	0.271	0.102
Cells	23,987	23,987	23,322	21,863	24,307
Panel B. Adding the average potential wage					
Relative potential wage	-0.056***	-0.057***	-0.036***	0.001	0.028***
	(0.009)	(0.010)	(0.009)	(0.026)	(0.008)
Average potential wage	0.093***	0.094***	0.026**	-0.135***	-0.008
	(0.010)	(0.010)	(0.010)	(0.026)	(0.008)

Notes: Specification, weights and clustering as in Table 1: equation (1) on 1980–2010 cells with Shenhav's relative potential wage; coefficients are per 10 percent increase; standard errors clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns 3 and 4 restrict each cell to ever-married and never-married women and are weighted by the number of such women; they condition on an outcome the relative wage moves and are descriptive. Column 5 is the share of mothers who have never married, weighted by the number of mothers.

Table 8: Carrying the Design to 2024

	1980–1990	1980–2000	1980–2010	1980–2024	2000–2024	2010–2024
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. First stage: observed log relative wage (CPS) on the relative potential wage						
Relative potential wage	1.278***	0.871***	0.593***	0.306***	0.119	0.001
	(0.237)	(0.228)	(0.191)	(0.102)	(0.099)	(0.139)
F-statistic	29.0	14.5	9.6	9.1	1.4	0.0
Panel B. Fertility on the extended relative potential wage (per 10% rise)						
Has at least one child	-0.002	-0.021*	-0.030***	-0.018***	-0.006	0.004
	(0.017)	(0.011)	(0.011)	(0.005)	(0.005)	(0.005)
Children per mother	0.030	-0.025	0.007	0.021*	0.019	-0.004
	(0.033)	(0.025)	(0.019)	(0.011)	(0.013)	(0.015)
Children per woman	0.008	-0.057*	-0.066***	-0.008	0.017	0.019
	(0.062)	(0.034)	(0.022)	(0.011)	(0.016)	(0.017)
Age at first birth	0.330**	0.288**	0.107*	0.261***	0.149***	0.121**
	(0.157)	(0.113)	(0.064)	(0.060)	(0.053)	(0.057)
Cells (has a child)	12,835	18,889	24,696	53,841	41,006	34,952

Notes: Panel A regresses the observed female-to-male log hourly wage gap in each matching market and year (March CPS) on the relative potential wage, following Shenhav [2021]: matching-market, education-by-year, race-by-year and state-by-year fixed effects and race-by-state trends, weighted by female population; the F-statistic is for the excluded instrument with state-clustered errors. Panel B estimates equation (1) on the relative potential wage rebuilt from the raw 1970 Census and March CPS and extended through 2024, using the 1980, 1990 and 2000 Censuses and the 2010, 2015, 2017, 2019, 2022 and 2024 ACS. Over 1980–2010 the rebuilt measure gives smaller coefficients than Shenhav's (Appendix Table A2). Coefficients in windows where Panel A shows no first stage cannot be interpreted as effects of relative wages. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix A Additional Tables

Table A1: Summary Statistics, 1980–2010

	1980	1990	2000	2010	All
Fertility outcomes (women aged 22–44)					
Has at least one child	0.676	0.651	0.649	0.620	0.648
	[0.211]	[0.179]	[0.172]	[0.215]	[0.196]
Children per mother	2.178	2.018	2.036	2.071	2.072
	[0.416]	[0.294]	[0.286]	[0.333]	[0.338]
Children per woman	1.472	1.314	1.322	1.284	1.343
	[0.659]	[0.493]	[0.479]	[0.577]	[0.556]
Age at first birth	22.548	23.479	24.179	24.238	23.653
	[1.879]	[2.229]	[2.708]	[2.950]	[2.581]
Share of mothers never married	0.037	0.073	0.117	0.181	0.103
	[0.079]	[0.121]	[0.149]	[0.195]	[0.152]
Potential wages (matching markets)					
Log relative potential wage	-0.258	-0.216	-0.214	-0.179	-0.215
	[0.048]	[0.048]	[0.054]	[0.062]	[0.060]
Log female potential wage	1.517	2.097	2.408	2.733	2.214
	[0.113]	[0.132]	[0.136]	[0.130]	[0.453]
Log male potential wage	1.775	2.312	2.622	2.911	2.429
	[0.105]	[0.135]	[0.152]	[0.162]	[0.432]
Women in sample	1,854,579	2,174,606	2,177,253	390,043	6,596,481
Cells	7,538	6,642	6,763	6,285	27,228

Notes: Means with standard deviations across cells in brackets, weighted by the female population of the cell (by the number of mothers for children per mother and the never-married share). 306 matching markets (state × education × race). Potential wages are from Shenhav [2021]. Data: IPUMS 1980, 1990 and 2000 Censuses and 2010 ACS.

Table A2: Shenhav's Instrument and the Rebuilt Instrument, 1980–2010

	(1) Shenhav (2021) instrument	(2) Rebuilt instrument	(3) Ratio (2)/(1)
Has at least one child	-0.043***	-0.030***	0.69
	(0.012)	(0.011)	
Children per mother	0.027	0.007	—
	(0.022)	(0.019)	
Children per woman	-0.071**	-0.066***	0.93
	(0.030)	(0.022)	
Age at first birth	0.337***	0.107*	0.32
	(0.088)	(0.064)	
Cells (has a child)	24,696	24,696	

Notes: Equation (1) on identical 1980–2010 cells, changing only the relative-wage variable. Column 1 uses the measure distributed by Shenhav [2021] (the main specification of the paper); column 2 the measure rebuilt from the raw 1970 Census and March CPS and extended to 2024, used in Section 8. Ratios are shown where column 1 is significant. Across 1,057 market-years the two measures correlate 0.991 in levels; after removing the fixed effects and trends of equation (1) the correlation is 0.776 and the slope of Shenhav's measure on the rebuilt one is 0.74, so classical measurement error alone implies coefficients roughly 26 percent smaller on the rebuilt measure. Standard errors clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix B Constructing the Potential Wage

Table A3 lists the 17 industry and 28 occupation groups used to construct the potential wage. They are taken from Shenhav [2021], who classifies occupations by broad IPUMS work-type categories and industries following Katz and Murphy [1992], with manufacturing split into three groups. Hourly wages in the Census and CPS follow her construction: imputed wages are dropped, top-coded annual earnings are multiplied by 1.5, hourly wages are capped at 1.5 times top-coded annual earnings divided by 1,400 hours, and the hourly wage is annual earnings divided by weeks worked times usual weekly hours. Wages are averaged with CPS weights multiplied by hours worked. The rebuilt measure used in Section 8 applies the same steps to the raw 1970 Census and the March CPS through 2025.

Table A3: Industry and Occupation Groupings

	Industry Groupings		**Occupation Groupings**
1.	Agriculture, forestry, and fishing	1.	Management
2.	Mining	2.	Engineers and scientists
3.	Construction	3.	Other technicians
4.	Low Tech Manufacturing	4.	Physicians/Nurses
5.	Basic Tech Manufacturing	5.	Health assistants
6.	High Tech Manufacturing	6.	Teachers and social workers
7.	Transportation	7.	Lawyers and judges
8.	Communication	8.	Entertainment
9.	Utilities	9.	Sales
10.	Wholesale Trade	10.	Administrative support
11.	Retail Trade	11.	Cleaning services
12.	Finance	12.	Other personal service
13.	Protective services	13.	Protective services
14.	Personal Services	14.	Food service
15.	Entertainment and Recreation	15.	Farm and forestry workers
16.	Professional Services	16.	Mechanical and electronic repair
17.	Public Administration	17.	Construction trades
		18.	Mining extraction
		19.	Metal or wood work or calibrators
		20.	Plant operator
		21.	Metal work operator
		22.	Textile work
		23.	Misc machine operator
		24.	Assemblers/fabricators
		25.	Vehicle operators
		26.	Construction, movers
		27.	Financial specialists
		28.	Management support

Source: Shenhav [2021].